

Tenstorrent LLM Bringup: Workflow, Correctness, and Performance

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Introduction

Tenstorrent’s **Blackhole P150** is a new ML accelerator (32 GB GDDR6, 110 Tensix cores, 1.39 MiB L1/core, 404 GB/s measured DRAM streaming). We brought up three modern LLMs on a 1×4 P150 mesh:

- **Qwen3.6-27B** — dense GQA, 4-way tensor-parallel.
- **Qwen3.6-35B-A3B** — MoE with Gated DeltaNet recurrence (~3 GB active params per token).
- **Gemma 4 12B** — hybrid sliding/global attention, dual head_dim (256/512), v_norm + per-layer scalar.

The first bringup (27B) took weeks; the third (Gemma 4 12B) walked from HF spec to HTTP chat in **36 hours**. The speedup is methodological — a forking recipe on stable scaffolding, not cleverer kernels.

The Bringup Workflow

Every model walks the same staging ladder. Each rung gates on cosine vs. an HF oracle, not on “it produces text”.

Stage	Adds	Gate
v0.0	HF CPU oracle (fp32, per-sub-op hooks)	dump exists
v0.1	L0 only on (1, 4) mesh	$\cos \geq 0.999$
v0.2	all N layers + lm_head	argmax @ pos 0
v0.3	paged KV + multi-step decode	argmax 0..7
v0.3.3	long context, needle haystack	top1 $\geq$ 90%
v0.4	trace capture (two-phase warmup)	100/100 traced == eager
v1	continuous batching $B=4$	3a/3b/3c gates
v2	HTTP wire-up	<code>curl /chat</code> works

HF oracle, not a sibling TT path. Per-sub-op forward hooks dump `.npy`; every TT change is gated against ground truth. Comparing TT-against-TT conflates “bug” with “different and better.”

Dev harness amortizes the bootstrap. A long-lived tmux’d Python process holds the model resident (~80 s for Gemma 4, ~14 min for 35B); file-trigger probes `importlib.reload` per fire. Iteration drops from one full bootstrap to ~30 s.

REUSE mandate. Every new file cites the prior file it forks in its commit message. After three bringups, ~80–90% of each new model’s CB module, dev harness, probes, and oracle is copied. Bug catalog is finite (~12 footguns); fixes are recognized in seconds.

Two-phase trace warmup. Compile all paths first (`enable_trace=False`), synchronize, then capture all traces back-to-back. Otherwise decode JIT between captures corrupts prefill trace memory → 99% CPU hang (tenstorrent/vlm#352).

Validation Hierarchy

Six of seven silent-failure classes pass at decode pos 0. The only gate that catches all of them is the **teacher-forced multi-step cosine + MAD ladder** against an HF oracle.

- **Teacher-forced cosine ladder** — per-position per-layer cos + MAD. Caught Gemma 4’s SDPA `scale=1.0` (invisible at pos 0; cos drops 0.99998 → 0.66 at pos 1) and Qwen3.6’s $y = x/\text{rms}(x) \cdot (1 + \gamma)$ zero-centered RMSNorm (a two-character fix lifted top-1 82/97 → 95/97, median cos 0.95 → 0.997).
- **Needle haystack** — random 8-char password at controlled depth. Passed Gemma 4 at $L=100/256/512$ (3/3 verbatim retrieval).
- **MAD, not just cosine.** $\cos(a, k \cdot a) = 1$ for any k ; missing Gemma 4’s `layer_scalar` passed at `cos=0.999975` with MAD 2.64×10^2 .

Correctness Stack

Three orthogonal gates, run per model on every change.

Gate	Qwen3.6-27B	35B-A3B	Gemma 4 12B
Cos ladder (pos ≤ 7)	PASS	7/8 PASS ($\cos_{L32} \geq 0.987$)	PASS
Argmax match @ pos 0	PASS	PASS	PASS
Multi-step token-for-token	PASS	5/6 (pos 4 = bf16 tie)	PASS
Needle haystack	$L=500$ PASS	~50% per trial (bf16)	3/3 ($L \leq 512$)
Multi-turn KV retention	PASS	3/3 PASS	PASS
Traced == Eager (100 toks)	PASS	PASS	100/100 PASS

The 35B free-run needle is *coherent failure, never gibberish* — a fundamental bf16 non-determinism floor, not a bug.

Performance Optimizations

Rule of thumb: kernel-time wins translate 1:1 in trace; dispatch-only fusions land at ~5–10% of their isolated eager gain. Predict trace ROI by asking “kernel work or dispatch?”

Optimization	Model	Δ tok/s	Why
Paged SDPA + B3 (HiFi2)	27B TP	+62%	kernel: $O(\text{MAX_POS}) \rightarrow O(1)$
Vocab-sharded lm_head	Gemma 4	+8%	248k/262k vocab shards 4×
Vocab-sharded lm_head	27B TP	+5.1%	on-device argmax: 600KB → 8B
A004 <code>core_grid=10x11</code>	35B-A3B	−30 ms/tok	ttnn auto-grid used 11/110 cores
A008 bf8 MoE expert weights	35B-A3B	−29 ms/tok	halves DRAM streaming
Owned fused decay/gate kernel	27B TP	+2.5%	10 ops → 1 (custom kernel)
<code>num_links=2</code> all_reduce	27B TP	+1.65%	2 links/axis on P150x4
A002 fused Q/K L2-norm	35B-A3B	−1.13 ms/tok	dispatch fusion only
DRAM-sharded MLP	27B	2.1×	slower Galaxy-70B recipe; wrong scale
Async <code>all_reduce_async</code>	27B TP	+4% overhead	no parallel CCL to overlap
fp32 KV cache	27B	dead end	paged SDPA rejects fp32
bf8 KV cache	27B	neutral	not memory-bound at $\leq 8k$

The 35B perf session stacked A002+A003+A004+A008 to take **141.79 → 81.16 ms/tok** (1.75×) in one day. The big lever is always kernel-time fusion or smarter weight streaming; trace already amortizes dispatch.

Throughput & Roofline

Decode at $B=1$ is weight-bandwidth-bound: ceiling $\approx \frac{\text{active GB/chip}}{404 \text{ GB/s}}$.

Model	Eager ms/tok	Traced ms/tok	tok/s	% of ceiling
Qwen3.6-27B (TP)	~253	77.0	12.93	12.93/60 = 22%
Qwen3.6-35B-A3B	480	81.16	12.32	12.32/270 = 4.6%
Gemma 4 12B	182.7	47.5	21.05	21.05/67.3 = 31%

Continuous batching — the largest single lever. Matmul weight reads amortize across B . HTTP stress ($B=32$ slots, paged SDPA, per-slot KV):

Clients	27B tok/s	vs 1c	Gemma 4 IT tok/s	vs 1c
1	5.36	1.00×	5.29	1.00×
8	40.87	7.63×	41.80	7.90×
16	79.59	14.85×	83.91	15.86×
32	156.59	156.59/5.36 = 29.2×	162.85	162.85/5.29 = 30.8×

Both models hit ~160 tok/s aggregate at 32 concurrent clients with near-perfectly-linear 29–31× scaling from one client, validating the CB design.

Multi-turn prefix cache (27B). Turn 2 = 9.03 s (172-tok prompt) vs. turn 1’s 19.26 s (105 tok): **6.3× speedup** on PC-hit turn. Gemma 4 PC still misses on chat-template re-renders (open bug).

Challenges

Six bug classes; each paid for in chip-hours and now in `research/model_bringup_recipe.md`.

- **ttnn view-decay.** `ttnn.slice` and `ttnn.reshape` return views; deallocating the source corrupts reads. Masked at pos 0 because `softmax(1) = 1` so `attn_out = V`. *Resolution:* never add `deallocate` calls the upstream pattern didn’t have.
- **Model-family semantic surprises.** Gemma 4 SDPA uses `scale=1.0`; Qwen3.6 RMSNorm is $(1 + \gamma)$ zero-centered; Gemma 4 attention has *three* per-head norms (q, k, v_norm) plus a per-layer `layer_scalar`. All invisible at pos 0. *Resolution:* multi-step teacher-forced + read HF.
- **bf16 chain non-determinism at $B>1$.** Every primitive bit-identical; 40-layer chain compounds ~1 ULP into argmax flips. *Resolution:* gate on per-slot independence, not exact tokens.
- **Silent perf regression — caught by audit subagent.** CB engine silently routed 27B through `manual` DN recurrence; first “fix” targeted `deltanet_*` (single-stream gate only). CB forward actually reads `cb_dn_recurrence_mode` which defaulted to `manual` via `getattr`. Two-line fix took $B=4$ HTTP 13.30 → $B=32 = 156.59$ tok/s. *Resolution:* audit subagent citing file:line.
- **Dev-harness PASS ≠ HTTP PASS.** 4/4 PASS in harness, first `/v1/chat/completions` crashed. *Resolution:* `curl` smoke every backend swap.
- **Chip changes overnight.** 35B drift cliff reproducible for days, vanished with no code change. P150 firmware v19.5.0 silently disabled 20 cores (140 → 120). *Resolution:* post-firmware regression runs.

What We Forked vs What We Built

Surface	Count	Origin
<code>ttnn.*</code> (SDPA, CCL, RMSNorm)	~1,570 calls	tt-metal
<code>ttnn.experimental.qwen36_*</code>	7 ops	ours (owned)
<code>owned_ops/ C++/HPP</code>	8,116 LOC	ours (Gated DN)
<code>serve/server_*.py</code>	16,206 LOC	wiring + HTTP
<code>cb/</code> (scheduler + harness)	11,383 LOC	ours (CB)

~225:1 TT:ours ratio, but the few originals gated correctness for each new model. Tenstorrent’s openness is the only reason a one-person project shipped three LLM families in a quarter.

Future Work

From the code-cleanup audit (18 items, 6 High).

- **Argmax-tail trace in HTTP** — closes the 156 → 376 tok/s gap on 27B $B=32$ (`cb_api.py` forces `sampling=True`).
- **Gemma 4 chat-template for prefix cache** — re-renders don’t byte-equal; port Qwen3.6’s patches for ~6× multi-turn speedup.
- **Gemma 4 P2/P3 distributed RMSNorm + paged SDPA globals** — projected +30–60%; closes the 31% → 60% roofline gap.
- **35B $B>1$ empty-slot poisoning (#162)** — unlocks 35B CB beyond `TT_CB_SLOTS=1`.
- **Delete dead manual DN path** — ~1.5K LOC; structurally broken (cos 0.08).

Conclusions

Three bringups, each faster (27B ~3 wks, 35B ~2 wks, Gemma 4 **36 hrs**). Compounding from *methodology*: staging ladder + HF oracle + dev harness + REUSE. CB delivers 156/163 **tok/s on 27B/Gemma 4** at $B=32$.